

Dynamics on aggregated connectomes in the browser: normalization, structural bias, and honest tuning lessons across two connectomes

Technical note v2, 2026-09-26. Code: [fruitfly/game/](#) @ commit [73da541](#) (MIT). Data: FlyWire FAFB v783 (CC BY-NC-SA 4.0) and MaleCNS v1.0 (CC BY 4.0). v2 adds a cross-dataset replication on MaleCNS (new §5): Finding 1 is strengthened, Finding 2 is partially refuted — both outcomes reported.

Repository: [fruitfly/game/](#) @ commit [73da541](#) · Demo: `python -m http.server 8000` → <http://localhost:8000> (?dataset=malecns 切换 MaleCNS)

中文摘要

本文记录浏览器游戏《电子果蝇·香蕉大作战》(Banana Quest) 在真实连接组上跑动力学时实测到的三个坑及其跨数据集复测结果，供连接组仿真社区参考。香蕉大作战用 FlyWire FAFB v783 全脑连接组 (139,255 神经元 / 2,698,236 条聚合连接，封装自 [snedea/flybrain](#)) 上的 LIF 脉冲网络驱动一只 2D

果蝇觅食，行为没有手工行为层。发现一：参考实现沿用的全局 max

权重归一化在本数据上传播失败——重尾突触计数分布 (最大 2405、中位数 8) 把典型权重压到阈值的约 1/2000，50 tick 后触角神经叶下游最大膜电压仅 0.688 (阈值 1.0)，中枢放电为零；改为 Shiu et al. 2024 式的按突触后神经元总输入归一化后， $\text{targetInput}=3.0$ 全链路可通、4.0

全脑点燃。发现二：下行神经元池按胞体 x 切半后存在结构性偏置——任一侧触角刺激右池都更活跃 (约 2100 对 1700 放电，有效差仅约 1.3%)，必须用慢速 EMA

基线的偏差而非原始放电率做转向读出。发现三 (方法论)：5 种子扫参的“冠军配置”在 10

个样本外种子上现形为运气，真正有效的改进来自关卡设计 (香蕉刷新距离环带 150–250px，首吃中位 27.3s) 而非读出参数注水。

v2 新增 (MaleCNS 跨数据集复测, §5)：在 MaleCNS v1.0 (176,422 神经元 / 6,287,749 条 ≥ 5

突触连接，含腹神经索) 上，发现一完整复现——重尾分布 (max 2591 / 中位 9) 与全局 max

归一化传播失败 (21,867 次放电全部困在感觉区、中枢 0、下游电压渐近 0.977

不过阈) 都是跨数据集现象；per-neuron 归一化的分层可达结构一致，但工作点漂移 (FlyWire 3.0 / MaleCNS 4.0)，且 ≥ 5 突触过滤使 MaleCNS 在 $t_i=4$ 时也不出现 FlyWire

式视觉系统点燃——结论是**必须逐数据集做传播测试**。发现二被部分修正：用 MaleCNS 原生 somaSide 池在充分驱动下侧向对比方向正确 (差约 8%)，v1

的“错误方向偏置”不复现——它更可能是粗粒度组定义 (GNG_DESC

混入上行神经元) 与坐标切半近似制造的伪影，而非数据集普遍性质；如实保留 v1

测量并公开自我修正，正是写这篇 note 的初衷。所有数字均可由文中命令复现。

Abstract

Banana Quest is a zero-dependency browser game in which a leaky integrate-and-fire (LIF) network running on a real connectome drives a 2D foraging fly, with no hand-crafted behavior layer. While building it we hit, measured, and worked around three problems that we believe are generic to the current wave of "run dynamics on a connectome" demos: (1) global-max weight normalization silently fails to propagate activity on aggregated connectomes with heavy-tailed synapse-count distributions; (2) coarse bilateral readout pools split by soma position can carry a structural bias larger than the stimulus-driven signal; and (3) small-seed parameter sweeps of high-variance behavioral metrics systematically overfit. v1 reported these on FlyWire FAFB v783 (139,255 neurons, 2,698,236 aggregated connections). **In this v2 we replicate the campaign on MaleCNS v1.0 (176,422 neurons, 6,287,749 connections at ≥ 5 synapses, including the ventral nerve cord): Finding 1 replicates completely — heavy-tailed weights (max 2591 / median 9) and global-max propagation failure (21,867 spikes trapped in sensory groups, zero central, subthreshold 0.977 asymptote) are cross-dataset phenomena; the per-neuron normalization structure replicates but the working point drifts (3.0 $\rightarrow$ 4.0), making per-dataset propagation tests mandatory; Finding 2 is partially refuted — with native somaSide pools and adequate drive, lateral contrast is correctly signed ($\sim 8\%$), so v1's wrong-direction bias was most likely an artifact of our own coarse group definitions, not a dataset-general property.** We report exact numbers, the fixes we adopted, and open questions for the connectome-simulation community.

1. Background

Banana Quest (电子果蝇·香蕉大作战) is an open-source, zero-build browser game. A virtual fly forages for bananas in a 2D arena. Olfactory input is injected as current into left/right pools of olfactory receptor neurons; spikes propagate through the connectome to descending neurons (GNG_DESC), whose pooled left/right firing-rate difference steers the fly, and whose total rate, scaled by a hunger variable, sets its speed. Feeding is gated by gustatory stimulation driving a SEZ_FEED + proboscis motor-neuron readout. There is deliberately no hand-tuned behavior layer: every turn, stop, and bite is conditioned on activity that actually traversed the 139k-neuron network.

The connectome ships as a compact binary (12.4 MB gzipped) packaged by the community project snedea/flybrain (MIT): FAFB v783 edges aggregated per neuron pair with signed weights (synapse counts signed by predicted transmitter class, GABA negative), plus a coarse 63-group functional annotation. Their reference simulator runs the full network in a Web Worker at ~ 10 ticks/s using a neuropil-gated dormant-group optimization; our simulation core (`src/sim-core.js`) is adapted from it and runs unchanged in both the browser worker and Node.js for testing.

Second dataset (v2). The game now also runs on MaleCNS v1.0 (`neuPrint male-cns:v1.0`; 176,422 neurons, 6,287,749 connections at ≥ 5 synapses carrying 90.3M synapses; the full male central nervous system including the VNC; CC BY 4.0), selected at runtime with `?dataset=malecns`. We take the FLYB v1 binary bundled by the fly-brain-minecraft project (MIT) and convert it with our own `tools/flyb_to_bin.py` into the same sim-core format — 26 functional groups with the same region encoding, and readout/stimulus pools filtered on MaleCNS's native annotations (somaSide, superclass, type) rather than v1's coordinate approximations. Every probe below runs through the identical `sim-core.js` on both datasets.

Why this note: the 2026 "flyslop" wave produced dozens of demos that drop a connectome into a spiking or rate model and tune weights until something moves. Almost none document what broke along the way. These pitfalls cost us real time, are easy to miss, and — as far as we can tell — are not written down anywhere. We write in the spirit of the awesome-fly disclaimer: "a moving fly... does not by itself demonstrate biological fidelity or learned behavior."

2. Finding 1 — Global-max normalization does not propagate on aggregated connectomes

2.1 Measurement

The reference implementation normalizes edge weights globally: $w \leftarrow w / \max|w| \times 0.15$. On this aggregated dataset the raw synapse-count weights are extremely heavy-tailed. A full scan of all 2,698,236 edges gives $\max|w| = 2405.0$, while a ~20k-edge sample gives median 8.0, p90 = 23.0, p99 = 83.0. Global-max normalization therefore maps the median edge to $\approx 0.15 \times 8/2405 \approx 5 \times 10^{-4}$ — roughly 1/2000 of the firing threshold (1.0). Propagation requires near-synchronous input budgets that biology-scale fan-in cannot deliver.

Empirically, driving the 926-neuron left food-odor ORN pool at sustained intensity 0.5 (firing every ~5 ticks) for 50 ticks produced 9,262 spikes, **all in sensory groups; central-brain spikes: 0**. The strongest single postsynaptic target of the pool receives a total signed weight of only 0.2825 spread over 372 presynaptic ORNs (largest single synapse 0.0036). The maximum membrane voltage reached in the downstream OLF_PN group climbed $0.299 \rightarrow 0.478 \rightarrow 0.585 \rightarrow 0.647 \rightarrow \mathbf{0.688}$ over ticks 10–50 and asymptotes below threshold. Signal dies in the antennal lobe.

2.2 Per-neuron normalization sweep

We replaced global scaling with Shiu et al. (2024)-style normalization — each postsynaptic neuron's total incoming $|w|$ is scaled to a constant `targetInput` — and swept it (100-tick probes driving the same ORN pool):

targetInput	result (100 ticks unless noted)
1.0	OLF_PN 287, OLF_LN 152, MB_KC 9, LH_APP 6, GNG_DESC 0 spikes
2.0	MB_KC 15,840, LH_APP 1,668, GENERIC_CENTRAL 4,602; GNG_DESC 0 (only 10 in 200 ticks at I=1.0)
3.0	GNG_DESC 3,810 in 200 ticks at I=1.0; no runaway over 500 ticks of maximal bilateral input (1,852–2,508 spikes/tick)
4.0	whole-brain ignition: GENERIC_CENTRAL 154,116 and VIS_ME 109,917 spikes per 100 ticks — the visual medulla fires massively with zero visual input

We ship `targetInput = 3.0` as default and keep `normalization: 'global-max'` as a switchable reference behavior. With it, the Node smoke test (0.5 intensity, 50 ticks) yields 47,106 total spikes of which 35,159 are in central regions, at ~1.5 ms/tick.

2.3 Discussion

Aggregation is the culprit multiplier: collapsing all synaptic contacts between a neuron pair into one scalar produces a few huge weights (a 2405-contact edge) that set the global scale for everyone else, and a threshold model then divides the world into "edges that matter" and "edges that don't" with nothing in between. Per-neuron input normalization is not a cosmetic choice on such data — it is the difference between a network that transmits and one that doesn't, and the viable window is narrow (3.0 works, 4.0 is epilepsy). Any demo that "just runs dynamics" on an aggregated connectome should publish its normalization and a propagation test; a silent network is the default outcome, not a bug in your stimulus.

3. Finding 2 — Structural bias in coarse descending readouts

3.1 Measurement

Our steering readout splits GNG_DESC (3,581 neurons) into left/right pools by soma x-coordinate median (1,791 / 1,790 neurons). Stimulating either antennal pool alone (I=1.0, 200 ticks, `targetInput 3.0`) gives:

stimulus	desc_left	desc_right	L/(L+R)
left antenna pool	1,704	2,106	0.447
right antenna pool	1,633	2,133	0.434

The right pool fires more under **either** unilateral stimulus (bias $\approx +24\%$ over the left pool), while the stimulus-dependent contrast between the two rows is only $\sim 1.3\%$. First descending spikes appear 9 ticks (~ 0.9 s at 10 Hz) after stimulus onset. Feeding the raw rates into a steering law $\text{turn} = k \cdot (R-L)/(R+L+\epsilon)$ would make the fly veer right forever regardless of where the banana is.

3.2 Baseline-deviation fix

We steer on the deviation from each pool's own slow baseline: $\text{turn} = k \cdot (R^{\wedge} - L^{\wedge}) / (R^{\wedge} + L^{\wedge} + \epsilon)$ with $R^{\wedge} = \max(0, R - \text{EMA}_{\tau}(R))$, $\tau = 5$ s. The EMA high-passes away the static bias and slow drift; the residual tracks the current gradient. Notably, a longer baseline ($\tau = 20\text{--}40$ s) performs worse in behavioral sweeps (eat rates fall to 40–80% in 7 of 8 swept configurations, and at $\tau = 40$ s path tortuosity rises to $\sim 17\text{--}23$): persistent wrong-sign deviations get "locked in" by the long memory, whereas the 5 s high-pass keeps the loop responsive.

3.3 Discussion

The bias is plausibly real structure, not a bug in our pools: the fly's olfactory pathway is largely bilateral, our x-median split is anatomically naive, and the dataset itself is a single individual whose two hemispheres may differ in proofreading completeness and annotation coverage. Schlegel et al. (2024) showed that more than half of all connection edges fail to replicate between FlyWire and hemibrain, so fine-grained lateral balance in a single specimen should not be trusted beyond a few percent anyway; Stürner et al. (2025) provide the comparative anatomy of descending/ascending neurons needed to do this properly. Practical rule for demo builders: **calibrate readout pools by baseline subtraction, and report the bias** — if your bilateral pools respond equally to both sides, your "steering" is cosmetic.

4. Finding 3 — Seed overfitting: a tuning-methodology lesson

4.1 The sweep winner that wasn't

Because steering is weak and noise-dominated, time-to-first-banana is a high-variance metric (in our runs, 12–166 s across seeds for identical parameters). We swept game-layer readout constants ($\text{turnGain} \times \text{smellSigma} \times \text{baseSpeed}$, then $\text{turnGain} \times \text{emaTau} \times \text{baseSpeed}$; 12 configs $\times$ 5 LCG-seeded runs $\times$ 180 simulated seconds each, brain reset between episodes, full tables in `game/README.md`). Two robust qualitative findings: raising `turnGain` monotonically hurts (it amplifies Poisson noise; path tortuosity climbs to ~ 18), and enlarging `emaTau` hurts (Section 3.2). The apparent champion, $\text{turnGain} = 1.8$, scored 100% eat rate, 26.4 s median, 3.00 mean score across the 5 sweep seeds.

On 10 fresh out-of-sample seeds it collapsed: 8/10 ate, median 57.7 s, mean 73.7 s, mean score 1.30 — **worse than the status-quo default on the same 10 seeds** (10/10, 49.6 s median, 2.70 score). The 5-seed "win" was seed luck.

4.2 What actually helped: level design

One control experiment is worth reporting as a negative result: capping banana spawn distance at 350 px changed nothing at all (identical metrics), because the fly spawns at the arena center of a 560 $\times$ 560 px arena where the farthest reachable point is only ≈ 362 px away — the constraint was geometrically vacuous. Capping spawns to a 150–250 px ring, however, moved the median time-to-first-banana from 49.6 s to **27.3 s** (10/10 eaten, tortuosity 8.47 — still clearly wandering search rather than a ≈ 1 straight-line beeline). A "fast search" variant (higher speed gain / max speed) that rewrote the rate $\rightarrow$ speed readout mapping improved score in one combination but degraded out-of-sample medians and was rejected.

4.3 Honest-tuning principle

The final shipped defaults change only the level-design constant (`bananaMaxDist = 250`). Every constant in the sensory→brain→readout mapping is unchanged, so the demo's core claim — "the brain's readout decides" — carries no hidden tuning water. The methodological lesson generalizes beyond games: **behavioral metrics with high stochastic variance must be validated out-of-sample; a small-seed sweep does not estimate the parameter effect, it estimates the seed set**. We kept both tables in the repo README precisely so the failure is auditable.

5. Cross-dataset replication on MaleCNS (new in v2)

Everything in §2–§4 was measured on one dataset. In v2 we re-ran the same probe battery on MaleCNS v1.0 through the identical simulation core. Full tables and commands are in `docs/malecns-probes.md`; the highlights and their consequences follow.

5.1 Preprocessing caveat (stated once, applies throughout)

The two packages differ upstream of our converter: the FlyWire binary keeps aggregated connections of any strength (2,698,236 edges), while the MaleCNS FLYB bundle is **thresholded at ≥5 synapses per connection** (6,287,749 connections — 24% of all connections, carrying 72% of all synapses) and drops autapses. MaleCNS is therefore a sparser but stronger graph, and it is male with a VNC, versus female brain-only for FlyWire. Cross-dataset differences below are reported with, not despite, these caveats.

5.2 Finding 1 — replicated, and strengthened

The weight distribution is heavy-tailed in essentially the same shape (raw synapse counts):

dataset	max	median	p90	p99
FlyWire v783	2405	8	23	83
MaleCNS v1.0	2591	9	28	93

Under `global-max` normalization, driving the 1,090-neuron left olfactory pool at 0.5 for 100 ticks produced **21,867 spikes, all in sensory groups; central-brain spikes: 0; motor spikes: 0**. The downstream `OLF_PN` group asymptotes at $\max V = 0.977$ — closer to threshold than FlyWire's 0.688, but still never crossing; `CB_INTRINSIC` reached only 0.089. Conclusion upgrade: propagation failure under `global-max` normalization is a **cross-dataset property of the aggregation × threshold-model × global-scaling combination**, not a FlyWire peculiarity. Two independent connectomes, two packagings, same silent network.

5.3 targetInput — structure replicates, the working point drifts

The per-neuron normalization sweep ($I=1.0$, 200 ticks, `olf_left`) reproduces the same layered reachability — 1.0 dies at the mushroom-body entrance, 2.0 ignites the mushroom body but not descending neurons, 3.0 first reaches them — but with two quantitative differences:

targetInput	MB_KC	DN (motor command)	VIS_OL (epilepsy indicator)
1.0	148	0	0
2.0	37,237	0	0
3.0	84,437	11	0
4.0	133,067	75	2

First, the olfactory→descending-neuron drive is **about two orders of magnitude weaker** than FlyWire's (11 vs 3,810 `GNG_DESC` spikes per 200 ticks at $t_i=3.0$). Two structural reasons: MaleCNS's `descending_neuron` superclass is a pure brain→VNC command population sitting several synapses from the olfactory entry, whereas FlyWire's "GNG_DESC" group mixes in directly sensory-driven neurons; and the ≥5-synapse filter removes weak cross-links. Second — an unexpected benefit of that filtering — **$t_i=4.0$ does not produce FlyWire-style visual ignition** (`VIS_OL`: 2 spikes, versus `VIS_ME`

109,917 spikes/100 ticks on FlyWire): the dense weak-edge cross-talk that fueled the seizure is simply absent. The viable working point is therefore dataset- and preprocessing-dependent (FlyWire 3.0, MaleCNS 4.0), and the practical advice to the community sharpens from "publish your normalization" to: **run a per-dataset propagation test and publish its working point — there is no universal constant.**

5.4 Finding 2 — partially refuted (an honest self-correction)

With MaleCNS's native somaSide pool split (no coordinate approximation) and adequate drive ($t_i=4.0$), the lateral contrast is **correctly signed**:

drive	stimulus	desc_left	desc_right	L/(L+R)
$t_i=3.0$, 200 ticks	left antenna pool	5	6	0.455
$t_i=3.0$, 200 ticks	right antenna pool	12	15	0.444
$t_i=4.0$, 400 ticks	left antenna pool	91	66	0.580
$t_i=4.0$, 400 ticks	right antenna pool	111	118	0.485

At $t_i=3.0$ the DN counts are too small (5–15 spikes per pool in 200 ticks; 10–34 in 400 ticks) for any bias statistic — itself a data point on how weak the drive is. At $t_i=4.0$, left stimulation favors the left pool and right stimulation the right pool, with ~8% contrast. **The wrong-direction structural bias of v1 does not replicate.** Where did it come from, then? A supplementary probe points at the group definition: MaleCNS ascending neurons (AN group) fired **0** times in 200 ticks under olfactory stimulation (versus 710 under bristle mechanosensory stimulation) — ascending neurons carry VNC→brain mechanosensory/proprioceptive signals, not olfaction. FlyWire's "GNG_DESC" group mixes ascending neurons in with descending ones, and its pools were split by a naive x-coordinate median; the bias was most likely manufactured by that combination of coarse grouping and coordinate approximation, not by a dataset-general circuit property.

We therefore restate Finding 2 more narrowly: **coarse functional group definitions can manufacture lateralization artifacts** — the pitfall is real, our v1 FlyWire measurements stand as the documented example, and baseline-subtraction remains the right defense — but the artifact is not a universal property of connectome data. Flagging this distinction is exactly why we wrote this note; a finding that survives contact with a second dataset and a finding that doesn't are equally worth publishing.

5.5 Biological sanity checks (small, but free)

- **DNp01 (giant fiber) does not respond to bristle stimulation:** 0 spikes in 20 ticks of strong mechanosensory-bristle drive — consistent with its known biology as a visually (looming-) triggered escape neuron. The game's escape path therefore keys on the descending-rate surge (bristle drives ascending/central activity strongly, 710 AN spikes in the same probe), not on DNp01.
- **The feeding chain works on MaleCNS:** gustatory pool at $I=1.2$ drives the MN9-class proboscis motor readout (16 cb_motor neurons) to 3 spikes in 30 ticks (first at tick 22) — at the game's feeding-gate threshold, and the headless closed loop on MaleCNS eats bananas for real (score 1–3 per 180 s across seeds, first eats 49–172 s; steering is legitimately "hard mode": DN rates of ~2–8 spikes/s at distance, rising to 34–42/s at close range).

6. Limitations and open questions

Limitations. Point LIF neurons without conductances, delays, neuromodulation, or plasticity; no gap junctions; transmitter signs from Eckstein et al. (2024) predictions (87% per-synapse / 94% per-neuron / 91% per-type accuracy) or MaleCNS's own consensus/predicted NT with an equivalent sign rule; two individuals (one female FAFB, one male MaleCNS) with different preprocessing (unthresholded vs ≥ 5 -synapse aggregation); FlyWire has no ventral nerve cord — locomotion there is a kinematic rendering of descending firing rates, and even on MaleCNS (which includes the VNC) we do not simulate real gait; game-layer kinematics (turn-and-slow, wall bounce, spawn ring, per-dataset stimulus gains) are acknowledged level design. None of this claims biological fidelity, let alone "uploading".

Open questions we would love the community to pick up (they dovetail with ongoing flybench-style validation efforts):

1. ~~Does global-max normalization failure replicate on MaleCNS?~~ **Answered in v2 (\$5.2): yes.** BANC remains untested — same one-command experiment.
2. ~~Is the rightward GNG_DESC bias dataset general?~~ **Answered in v2 (\$5.4): no — it was a group-definition artifact.** The follow-up that remains interesting: which named DN types (e.g. the turn/halt DNs characterized in the literature) restore steering contrast without any baseline subtraction, on either dataset?
3. Why does olfactory→DN→leg-motor drive stall at the DN stage on MaleCNS (leg motor pools fired 0 times in 400-tick probes even at $t_i=4.0$)? Longer timescales, neuromodulatory gating, or the ≥ 5 -synapse filter amputating the weak tail of DN→MN synapses?

7. Reproducibility

All numbers in this note were produced on this repository at commit 73da541 (Node v24, Python 3.13, Windows; runs are CPU-only and complete in minutes). v1 numbers were produced at commit d92807b with the same commands and are bit-stable across reruns we checked.

number(s)	command
139,255 neurons / 2,698,236 edges; propagation 47,106 spikes / 35,159 central; ms/tick	<code>node game/test/sim.test.mjs</code>
weight distribution (max 2405, median 8, p90 23, p99 83); pool sizes	<code>python game/tools/prepare_pools.py</code>
normalization failure baseline ($V=0.688$, 0 central spikes)	<code>node game/test/sim.test.mjs</code> with normalization: 'global-max' in <code>src/sim-core.js</code> (LIFSim.fromBuffer opts)
targetInput sweep, bias table, latency (FlyWire)	inline probes against <code>src/sim-core.js</code> (same code path as the smoke test; see note history)
closed-loop validation (first eat 29.6 s, tortuosity 5.34, score 2/120 s)	<code>node game/tools/headless_run.mjs 120</code>
sweep round 1 ($\text{turnGain} \times \text{smellSigma} \times \text{baseSpeed}$)	<code>node game/tools/tune_sweep.mjs</code>
sweep round 2 ($\text{turnGain} \times \text{emaTau} \times \text{baseSpeed}$)	<code>node game/tools/tune_sweep.mjs '{"turnGain":[1.8,2.6], "emaTau":[5,20,40], "baseSpeed":[6,12]}'</code>
out-of-sample & level-design comparisons	<code>runEpisode()</code> API in <code>game/tools/harness.mjs</code> (seeds 101–110)
MaleCNS three-file conversion (176,422 neurons / 6,287,749 edges; weight stats max 2591 / median 9 / p90 28 / p99 93)	<code>python game/tools/flyb_to_bin.py</code>
MaleCNS probes 1–4 (all \$5 tables)	<code>node game/tools/probe_malecns.mjs</code>
MaleCNS feeding/escape/ascending supplementary probes	inline scripts driving LIFSim + <code>data/pools_malecns.json</code> at $t_i=4.0$ (transcribed in <code>docs/malecns-probes.md</code>)

number(s)	command
MaleCNS closed loop (score 1, first eat 104.3 s, tortuosity 3.36)	node game/tools/headless_run.mjs 180 '{}' 7 malecns

The MaleCNS chain is: neuPrint male-cns:v1.0 (CC BY 4.0, Berg et al. 2026) → FLYB v1 binary bundled by fly-brain-minecraft (MIT; format spec in its PROVENANCE.md §4) → our tools/flyb_to_bin.py converter → game/data/connectome-malecns.bin.gz + neuron_meta_malecns.json + pools_malecns.json. The game selects it via `http://localhost:8000/?dataset=malecns`.

Seeded runs use an injectable LCG replacing `Math.random`; episodes call `sim.reset()` between runs so one process reuses a single parsed brain. Headless runs are not real-time throttled: 180 simulated seconds complete in a few wall-clock seconds.

References

- Bates, A. S., Phelps, J. S., et al. (2026). Distributed control circuits across a brain-and-cord connectome. Nature. <https://doi.org/10.1038/s41586-026-10735-w>
- brandoncho369/flybench (2026). Behavioural benchmark for whole-brain fruit fly connectome simulations: pre-registered, cited behavioural tasks on FlyWire v783 and MaleCNS with shuffled-wiring controls and a reference LIF implementation. <https://github.com/brandoncho369/flybench> (MIT)
- blendi-remade/fly-brain-minecraft (2026). Minecraft mod driven by the MaleCNS connectome; source of the FLYB v1 binary packaging of neuPrint male-cns:v1.0 used here (format spec and full transformation log in its PROVENANCE.md). <https://github.com/blendi-remade/fly-brain-minecraft> (MIT)
- Beiran, M. & Litwin-Kumar, A. (2025). Nature Neuroscience 28: 2561–2574 (连接组只部分约束动力学) . Code: <https://github.com/emebeiran/connconstr> ; data: <https://zenodo.org/records/16618353>
- Berg, S., Beckett, I., et al. (2026). Sexual dimorphism in the complete connectome of the Drosophila male central nervous system (MaleCNS). Cell. doi:10.1016/j.cell.2026.08.015 ; preprint: <https://www.biorxiv.org/content/10.1101/2025.10.09.680999v1>
- cobanov/awesome-fly (2026). Community index of fly-brain demos, with fidelity disclaimers. <https://github.com/cobanov/awesome-fly>
- Dorkenwald, S., Matsliah, A., Sterling, A. R., et al. (2024). Neuronal wiring diagram of an adult brain. Nature 634: 124–138. <https://doi.org/10.1038/s41586-024-07558-y>
- Eckstein, N., Bates, A. S., Champion, A., et al. (2024). Neurotransmitter classification from electron microscopy images at synaptic sites in Drosophila melanogaster. Cell 187: 2574–2594.e23. <https://doi.org/10.1016/j.cell.2024.03.016>
- Schlegel, P., Yin, Y., Bates, A. S., et al. (2024). Whole-brain annotation and multi-connectome cell typing of Drosophila. Nature 634: 139–152. <https://doi.org/10.1038/s41586-024-07686-5>
- Shiu, P. K., Sterne, G. R., Spiller, N., et al. (2024). A Drosophila computational brain model reveals sensorimotor processing. Nature 634: 210–219. <https://doi.org/10.1038/s41586-024-07763-9>
- snedea/flybrain (2026). Browser-based LIF simulation of the FlyWire connectome (data packaging & reference simulator). <https://github.com/snedea/flybrain>
- Stürner, T., et al. (2025). Nature 643: 158–172 (下行/上行神经元的比较连接组学: FlyWire + MANC + FANC) . <https://doi.org/10.1038/s41586-025-08925-z>
- Wang-Chen, S., Stimpfling, A., Lam, K., et al. (2024). NeuroMechFly v2: simulating embodied sensorimotor control in adult Drosophila. Nature Methods 21: 2353–2362. <https://doi.org/10.1038/s41592-024-02497-y>
- Zheng, Z., Lauritzen, J. S., Perlman, E., et al. (2018). A complete electron microscopy volume of the brain of adult Drosophila melanogaster. Cell 174: 730–743.e22. <https://doi.org/10.1016/j.cell.2018.06.019>

Community context: this note was written against the backdrop of the 2026 "flyslop" demo wave (indexed at awesome-fly and Fly Brain Hub, <https://fly-brain-hub.vercel.app/>) and community validation efforts such as flybench.